

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 8: Appendix

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline that produced the numerics in Chapter 4: data sources, processing stages, default parameters, proxy formulas, and a glossary. All quantities cited here also appear in the main-text tables and figures; no external files are required to interpret the claims.

8.1.1 Pipeline stages

The evaluation proceeds in three stages on Arbitrum One over the observation window 2025-12-01 to 2026-05-31 (UTC, inclusive).

1. **Trading-success labels:** Decode GMX V2 `PositionDecrease` events from the public Big-Query log table, recover wallet identity, realized PnL, position-size change, and liquidation flags, and retain wallets with at least three closes as the matched cohort ($n = 5,521$; 365,488 decoded closes).
2. **Reputation graphs and scores:** Extract ERC-20 `Approval` and `Transfer` logs for cohort wallets in monthly batches (per-query dry-run and a 150 GiB abort cap). Build the latest-allowance endorsement graph and the time-decayed transfer graph, then compute `EndorseRank`, `AWP`, and supplementary GF-PR scores by power iteration (damping $d = 0.85$, tolerance 10^{-8} , at most 300 iterations).
3. **Proxies, alignment, and benchmarks:** Compute the five proxy families defined in Chapter 3, report Spearman ρ and Kendall τ against each score vector, time PageRank solves on pre-built edges (three-run averages), and run robustness sweeps over damping, top tokens, and sample size.

Offline synthetic checks use the same scoring and alignment logic on fixture graphs (no cloud access) to validate implementation before real extraction. Real extraction requires Google Cloud credentials and a billing project; scans target only the public Arbitrum One log dataset.

8.1.2 BigQuery extraction design

ERC-20 Approval and Transfer events are identified by their standard topic-0 hashes and restricted to logs whose owner/spender or from/to address lies in the evaluation wallet list, so scan cost scales with the cohort rather than the full chain. Monthly batching keeps each query under the operational byte cap. GMX V2 events are read from the EventEmitter contract at 0xC8ee91A54287DB53897056e12D9819156D3822Fb and decoded to key–value fields for account, basePnlUsd, sizeDeltaUsd, and liquidation flags. Representative SQL patterns appear in Appendix 8.2. Extraction records (bytes processed, row counts, timestamps) are retained in the study’s acquisition log for auditability; the dissertation reports the resulting cohort sizes and edge counts rather than those operational logs.

8.1.3 Scaling tables

Table 8.1 reports in-cohort runtime scaling for $n \leq 5,521$. Robustness tables in Chapter 4 use the same solver settings and cohort definitions as the primary evaluation.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort).

n	ER (s)	AWP (s)	Speedup	ER $ E $	AWP $ E $
1000	0.393	3.215	$8.2\times$	2584	23058
2000	0.729	6.794	$9.3\times$	5147	47031
4000	2.075	18.031	$8.7\times$	10557	97105
5521	2.105	18.711	$8.9\times$	14727	137087

8.1.4 Credit-delegation extract (deferred)

Aave V3 BorrowAllowanceDelegated events were not extracted in this study. Any exploratory ranking that depends on credit-delegation edges is therefore undefined on the present sample and is reported as non-interpretable ($\tau = \text{---}$) in the archived Aave-oriented matrix (Appendix 8.4). Extending the study to delegation-based reputation would require a dedicated extract of those events and a fresh alignment evaluation; it is left to future work.

8.1.5 Supplementary alignment summary table

Table 8.2 restates family-level mean τ from Table 4.7 in columnar form (EndorseRank / AWP / GF-PR / highest-alignment label). The main narrative uses Table 4.7; the columnar form is retained here for side-by-side reading.

Table 8.2: Mean Kendall τ by proxy family: EndorseRank, AWP, and GF-PR ($n = 5521$).

Family	EndorseRank	AWP	GF-PR	Highest
Transfer	0.333	0.534	0.246	AWP
Allowance	0.355	-0.030	-0.041	EndorseRank
Liq.	0.093	0.113	0.132	GF-PR
Inv.risk	0.035	0.107	0.376	GF-PR
Sybil	0.233	0.286	0.093	AWP
GMX	0.096	0.179	0.583	GF-PR

8.1.6 Parameter summary

8.1.7 Proxy operational definitions (reference)

The following formulas match Chapter 3.

- **Transfer:** $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.
- **Allowance:** $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_o w_a(o, w)$ with w as spender.

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations	300	Power-iteration cap
Logistic k	0.01	AWP decay slope
Logistic t_0	180 days	AWP decay midpoint
Min. GMX closes	3	Cohort eligibility
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
GF-PR loss $\varepsilon_{\text{loss}}$	0.1	Ordinary loss edge boost
GF-PR liquidation ε_{liq}	1.0	Liquidation edge boost
Expanded pool subsample seed	benchmark-tier2-v1	Deterministic SHA256 seed

- **Inverse risk:** $\text{loss_avoidance}(w) = -\sum_c \min(\text{PnL}_c, 0)$; $\text{non_loss_close_rate}(w) = 1 - |\{c : \text{PnL}_c < 0 \wedge \neg \text{liq}_c\}|/|\mathcal{C}(w)|$; $\text{worst_close_pnl_score}(w) = -\min_c \text{PnL}_c$.
- **Sybil-adjusted stability:** $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}|/|T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.
- **Trading success:** $\text{close_success_count}(w) = |\{c : \text{PnL}_c > 0 \wedge \neg \text{liq}_c\}|$; $\text{realized_gain_proxy}(w) = \sum_c \max(\text{PnL}_c, 0)$; $\text{close_success_rate}(w) = \text{close_success_count}(w)/|\mathcal{C}(w)|$.

8.1.8 Glossary

Allowance ERC-20 authorization for a spender to transfer tokens on behalf of an owner.

Alignment Rank agreement between reputation scores and a validation proxy (ρ, τ) .

Construct overlap Shared data source between a ranking method and a proxy, inflating alignment.

Endorsement graph Directed graph of latest allowances (EndorseRank input).

Matched cohort Primary sample of $n = 5,521$ wallets meeting eligibility rules.

Expanded wallet pool Larger address set ($N = 31,612$) for runtime scaling.

PageRank Spectral ranking by stationary distribution of a damped random walk.

Proxy Observable wallet metric used for domain-intuition validation.

Teleport Uniform jump in PageRank with probability $1 - d$.

8.2 SQL Templates, Configuration, and Extended Notes

This appendix collects representative SQL patterns, configuration keys, and extended methodological notes that support reproduction without cluttering the main chapters.

8.2.1 Approval log extraction pattern

ERC-20 Approval events are identified by topic 0 equal to the keccak256 hash of Approval(address,address,uint256). A simplified monthly query pattern is:

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS owner_topic,
    topics[SAFE_OFFSET(2)] AS spender_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
```

```

    OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
    ↪ UNNEST(@wallet_list)
);

```

Wallet filtering via @wallet_list keeps scans proportional to the cohort rather than the full chain. Monthly batching respects per-query byte caps documented in the study’s acquisition record.

8.2.2 Transfer log extraction pattern

Transfer events use topic 0 for Transfer(address,address,uint256):

```

SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS from_topic,
    topics[SAFE_OFFSET(2)] AS to_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
    );

```

Post-processing applies token decimals, builds time-decayed edge weights, and restricts nodes to the evaluation sample before PageRank.

8.2.3 GMX PositionDecrease extraction pattern

GMX V2 events are read from the `EventEmitter` contract. The pipeline filters on the emitter address `0xC8ee91A54287DB53897056e12D9819156D3822Fb` and decodes key-value fields to recover `account`, `basePnlUsd`, `sizeDeltaUsd`, and liquidation flags. Only `PositionDecrease` rows enter proxy computation. Wallets with fewer than three closes are excluded from the matched cohort.

8.2.4 Configuration keys

The pipeline’s parameters are collected in a single versioned configuration file. Salient keys include:

- observation window start and end timestamps;
- PageRank damping, tolerance, and maximum iterations;
- logistic decay k and t_0 ;
- GMX emitter address and topic filters;
- minimum closes for cohort eligibility ($m = 3$);
- benchmark repetition count and subsample seeds;
- output locations for intermediate artifacts and LaTeX fragments.

Changing damping or decay parameters requires re-running the full evaluation—including robustness sweeps and table export—and regenerating figures from the same artifacts, so that reported numerics remain consistent with the pipeline’s machine-readable summary.

8.2.5 Complexity notes

Let n be the number of nodes and $m = |E|$ the number of edges. Building latest-allowance edges is $O(R_a \log R_a)$ if R_a approval rows are sorted by timestamp, or $O(R_a)$ expected time with a hash map keyed by (token, owner, spender). Building transfer edges is $O(R_t)$ over transfer rows. PageRank is $O(mI)$ for I iterations. Empirically $I \approx 37$ for EndorseRank and $I \approx 100$ for AWP on the matched cohort, and $m_{\text{approve}} \ll m_{\text{transfer}}$, so both factors favor EndorseRank.

Memory is dominated by edge lists and score vectors. Peak solve memory of 4.6 MB versus 38.2 MB (Table 4.2) is consistent with the edge-count ratio 14,727 : 137,087.

8.2.6 Interpretation checklist for readers

When reading Chapter 4 tables, the following checklist reduces common misreadings:

1. Identify whether a row is a *social* method (EndorseRank, AWP) or the supplementary GF-PR reference.
2. Match each family to its intended construct (allowance vs. transfer vs. outcome).
3. Prefer relative comparisons on the same cohort over absolute τ thresholds.
4. Treat GF-PR highs on trading-success and inverse-risk as construct-overlap diagnostics (SRQ1).
5. Read runtime tables as PageRank solve time on pre-built edges, not end-to-end ETL time.
6. Confirm robustness tables before generalizing a point estimate.

8.2.7 Archived extended baselines

Seven-method and six-Aave alignment matrices are archived alongside the evaluation pipeline, together with corresponding CSV matrices for spreadsheet inspection. They are excluded from the main narrative because several variants require additional event types (e.g., credit delegation) or introduce constructs outside the primary EndorseRank–AWP question. Researchers extending this work may re-enable exports via a dedicated export utility with documented presets.

8.2.8 Limitations reprise

Appendix material does not expand the empirical claims beyond Chapter 4. In particular, SQL templates do not imply that other chains were queried, and archived baselines do not constitute peer comparisons in this dissertation. The authoritative numerics remain those in the pipeline’s machine-readable summary artifact for the matched cohort ($n = 5,521$) and expanded wallet pool ($N = 31,612$).

8.3 Elementary Properties of PageRank and Rank Correlation

This appendix records elementary properties used implicitly in Chapters 2–4. The material is standard ^{[36],[40],[49],[54]} and is included for self-contained reading.

8.3.1 Existence and uniqueness of PageRank

Let P be a row-stochastic matrix on n nodes, possibly after dangling-node repair as in Equation (2.1). For $d \in (0, 1)$, the Google matrix

$$G_{\text{PR}} = dP^\top + \frac{1-d}{n}\mathbf{1}\mathbf{1}^\top$$

is positive and column-stochastic (up to the transpose convention in Equation (2.2)). By the Perron–Frobenius theorem for positive matrices, G_{PR} has a unique eigenvalue of modulus one with a strictly positive eigenvector π normalized so that $\sum_i \pi_i = 1$. Hence PageRank scores exist and are unique for any directed graph once damping and dangling-node rules are fixed ^[40].

This uniqueness underpins comparisons between EndorseRank and AWP: given fixed parameters, each method induces a single score vector on a given edge set. Differences between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ are therefore attributable to edge semantics and weights, not to solver multiplicity.

8.3.2 Continuity in edge weights

PageRank is a continuous function of the entries of P for fixed $d \in (0, 1)$. Small perturbations of allowance or transfer weights therefore induce small perturbations of scores in the $\|\cdot\|_1$ norm, with Lipschitz constants depending on d and n ^[40]. Continuity justifies robustness checks that vary damping or restrict to top tokens: if alignment were knife-edge fragile, small hyperparameter changes would scramble family-level τ . Empirically, EndorseRank allowance τ remains ≈ 0.355 across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.4), consistent with a stable ranking geometry.

8.3.3 Sparsity and per-iteration cost

A sparse matrix–vector product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations for m nonzero edges. Power iteration therefore costs $O(mI)$ for I iterations. On the matched cohort, $m_{\text{ER}} = 14,727$ and $m_{\text{AWP}} = 137,087$, so even at identical I the transfer method would be roughly $9.3\times$ more expensive per iteration. Observed iteration counts ($I_{\text{ER}} \approx 37$, $I_{\text{AWP}} \approx 100$) widen the gap, matching the measured wall-time ratio $\approx 8.9\times$ (Table 4.2).

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation applied to ranks. It is invariant to strictly increasing transformations of either variable and lies in $[-1, 1]$. Perfect monotone agreement yields $\rho = 1$; perfect reverse agreement yields $\rho = -1$. Ties require mid-rank adjustments; the evaluation pipeline uses standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association is more appropriate than Pearson correlation on raw values ^[16].

8.3.5 Properties of Kendall’s τ

Kendall’s τ equals the normalized difference between concordant and discordant pairwise orderings. It admits a probabilistic reading: for two random distinct wallets, τ is the probability their relative order agrees under both rankings, minus the probability it disagrees ^[36]. This reading is use-

ful when communicating results to practitioners: $\tau = 0.355$ means a clear but imperfect tendency for allowance-proxy orderings to agree with EndorseRank orderings.

For large n , τ and ρ are often similar in sign and magnitude but not identical. This dissertation reports both, following Do et al.^[16], and uses family-mean τ as the primary summary.

8.3.6 Construct validity note

Cronbach and Meehl^[14] distinguish criterion-related validity from construct validity. In the absence of a gold-standard default label, this dissertation pursues a construct-oriented strategy: each method is tested against proxies that operationalize its intended meaning (allowance receipt for EndorseRank; inbound transfer activity for AWP) and against secondary families that probe neighboring constructs (risk, stability, trading success). High within-construct τ supports interpretation; low cross-construct τ is not automatically failure—it may indicate discriminant validity. GF-PR’s extreme trading-success τ is a cautionary case of *contaminated* criteria, where the criterion shares inputs with the measure (SRQ1).

8.3.7 Numerical summary sheet

For quick reference, the primary matched-cohort numerics are:

Quantity	Value
Matched wallets n	5,521
GMX closes (decoded)	365,488
EndorseRank edges	14,727
AWP edges	137,087
EndorseRank runtime	2.105 s
AWP runtime	18.711 s
Runtime ratio (AWP/ER)	$\approx 8.9\times$
Inter-method τ	0.316
ER allowance mean τ	0.355
AWP transfer mean τ	0.534
ER trading-success mean τ	0.096
AWP trading-success mean τ	0.179
GF-PR trading-success mean τ	0.583
GF-PR inverse-risk mean τ	0.376
Expanded pool N	31,612
Speedup at $n = 10,000$	$\approx 10.4\times$
Speedup at $N = 31,612$	$\approx 9.4\times$
Observation window	2025-12-01 – 2026-05-31

These values are transcribed from the pipeline’s machine-readable evaluation summary and match the exported result tables.

8.4 Archived Extended Baselines

During exploratory analysis, additional PageRank variants were scored on the same matched cohort and proxy families. They are archived here for completeness and future work. They are *not* part of the primary EndorseRank–AWP research questions and should not be read as peer claims in the

main narrative.

8.4.1 Seven-method alignment matrix

Table 8.4 reports family-level mean Kendall τ for seven methods, including LP-PR, CW-AWP, LF-PR, and RiskProp-style variants implemented in the evaluation codebase. EndorseRank remains the allowance-family leader; AWP (and closely related transfer variants) lead on transfer; GF-PR leads on inverse-risk and GMX success by construction.

Table 8.4: Mean Kendall τ by proxy family across 7 PageRank methods ($n = 5521$). Bold = column maximum. Runtime = mean PageRank wall time (s, 3 repeats). Edge counts: liq-pr=45, borrow-pr=484, repay-pr=787, borrow-pr-pool=262, repay-pr-pool=232, delegation-pr=0.

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX	Runtime (s)
AWP	0.534	-0.030	0.113	0.107	0.286	0.179	22.757
EndorseRank	0.333	0.355	0.093	0.035	0.233	0.096	2.544
GF-PR	0.246	-0.041	0.132	0.376	0.093	0.583	1.879
LP-PR	0.529	-0.027	0.136	0.098	0.280	0.196	27.630
CW-AWP	0.504	-0.013	0.128	0.114	0.253	0.204	20.364
LF-PR	0.197	-0.031	0.045	0.017	0.102	0.032	0.113
RiskProp	0.534	-0.030	0.113	0.107	0.286	0.179	24.301

Interpretation guidelines:

- Methods built on transfer-like edges cluster near AWP on transfer and Sybil families.
- Methods with very small edge counts (e.g., liquidation-oriented graphs with dozens of edges) can produce unstable or degenerate rankings; their runtimes may be low simply because m is tiny.
- RiskProp-style rows that mirror AWP numerically indicate shared transfer structure rather than independent discovery.
- GF-PR’s dominance on outcome families remains a construct-overlap warning, identical to Chapter 4.

8.4.2 Six-method Aave-oriented matrix

Table 8.5 reports an Aave-oriented preset. Several lending-event graphs are sparse on Arbitrum within the study window; delegation-based rows may be degenerate when BorrowAllowanceDelegated events were not extracted (Appendix 8.1).

Table 8.5: Mean Kendall τ by proxy family across 6 PageRank methods ($n = 5521$). Bold = column maximum. Runtime = mean PageRank wall time (s, 3 repeats). Aave W→W edges: LiqCall (liquidator→user), Borrow (initiator→onBehalfOf), Repay (repayer→user), Delegation (delegator→delegatee). Credit Delegation extract deferred when parquet absent— τ may be degenerate. Edge counts: liq-pr=45, borrow-pr=484, repay-pr=787, borrow-pr-pool=262, repay-pr-pool=232, delegation-pr=0.

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX	Runtime (s)
AWP	0.534	-0.030	0.113	0.107	0.286	0.179	22.757
EndorseRank	0.333	0.355	0.093	0.035	0.233	0.096	2.544
LiqCall-PR	0.053	-0.012	-0.008	0.026	0.025	0.030	0.011
Borrow-PR	0.113	-0.019	0.020	0.014	0.055	0.023	0.083
Repay-PR	0.143	-0.024	0.016	0.022	0.068	0.026	0.131
Delegation-PR	—	—	—	—	—	—	0.000

These matrices motivate future work on credit-delegation edges and cross-protocol lending labels, but they do not alter Claims C1–C4 or SC1.

8.4.3 Why the main narrative uses three methods

Focusing the dissertation on EndorseRank, AWP, and supplementary GF-PR serves three goals:

1. **Clarity of contribution:** The novel construct is allowance-based endorsement. Comparing it to the established transfer baseline ^[16] isolates the research question.
2. **Avoiding baseline sprawl:** Additional variants that share transfer edges with AWP add rows without adding independent constructs.
3. **Honest use of GF-PR:** Keeping GF-PR supplementary prevents readers from treating an outcome-native ceiling as a social reputation competitor.

Researchers who need broader baselines can regenerate the archive tables from the open evaluation

pipeline using the seven-method and six-Aave presets.

8.4.4 Reproducing archive tables

Archive tables are regenerated by the evaluation pipeline’s table export with seven-method and six-Aave presets. CSV copies are recorded alongside the LaTeX fragments for spreadsheet inspection.

8.4.5 Notes on archived method identifiers

The following identifiers appear in archive tables and CSV exports. They document exploratory implementations in the evaluation codebase.

LP-PR A transfer-graph variant with alternative loss-penalized or liquidity-oriented weighting explored during development. Empirically it tracks AWP closely on transfer families (Table 8.4), indicating shared payment-flow structure.

CW-AWP A counterparty-weighted AWP variant that down-weights repeated transfers between the same pair. It slightly reduces transfer τ relative to AWP while remaining in the same construct family.

LF-PR A low-frequency or lightly filtered graph with very small edge counts and near-zero runtimes; rankings are not comparable to full AWP and are retained only to document a failed sparse baseline.

RiskProp A risk-propagation style ranking on transfer-like edges inspired by Lin et al.^[44]. On this cohort it numerically mirrors AWP on several families, suggesting that the implemented risk signal did not diverge strongly from transfer centrality under the study’s parameterization.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs built from liquidation calls, borrows, and repays when those events are present. On Arbitrum in the study window, edge counts are small (tens to hundreds), so alignment is weak and runtimes are negligible.

Delegation-PR Intended to use credit-delegation allowances; with zero delegation edges extracted, scores are degenerate and τ is not interpretable.

These notes exist so that future researchers can decide which variants deserve full re-implementation rather than treating archive rows as peer-reviewed baselines.

8.4.6 Decision log: what stayed in the main text

The decision to keep only EndorseRank, AWP, and supplementary GF-PR in Chapters 4–5 was made after inspecting the seven-method and six-Aave matrices. Criteria were:

1. **Construct independence:** A baseline must implement a distinct edge semantics, not a minor reweighting of transfers.
2. **Data adequacy:** Edge counts must be large enough for stable PageRank on the matched cohort.
3. **Narrative focus:** The dissertation’s contribution is allowance-based endorsement, not a survey of all possible PageRank variants.
4. **Evaluation honesty:** Outcome-native methods remain supplementary diagnostics.

Under these criteria, EndorseRank (allowance), AWP (transfer), and GF-PR (outcome-native ceiling) form a minimal sufficient set. Archive tables remain available for readers who want the broader landscape without diluting the primary claims.

8.5 Claim-to-Evidence Map

This appendix maps each hypothesis, research question, and empirical claim to the specific evidence in Chapter 4. It is intended as a navigation aid for examiners and replicators.

8.5.1 H1 / RQ1 — Rank divergence

- **Statement:** EndorseRank and AWP yield distinct wallet orderings on the same cohort.

- **Evidence:** Inter-method Kendall $\tau = 0.316$ between score vectors (Section 4.6); rank scatter in Figure 4.8; score distributions in Figure 4.7.
- **Interpretation band:** Weak-to-moderate agreement: related but not interchangeable rankings.

8.5.2 H2 / RQ2 — Construct-specific alignment

- **Statement:** Social methods differ in which proxy families they align with.
- **Evidence:** Table 4.7; per-family Tables 4.8–4.12; heatmap Figure 4.6.
- **Winners:** EndorseRank on allowance ($\tau = 0.355$ vs. -0.030); AWP on transfer (0.534 vs. 0.333) and trading success among social methods (0.179 vs. 0.096).

8.5.3 H3 / RQ3 — Computational efficiency

- **Statement:** EndorseRank is faster than AWP at equal n .
- **Evidence:** Table 4.2 (2.105 s vs. 18.711 s, $\approx 8.9\times$); Table 4.3 ($\approx 10.4\times$ at $n = 10,000$; $\approx 9.4\times$ at $N = 31,612$); Figure 4.4; edge counts $14,727$ vs. $137,087$.

8.5.4 RQ4 — Validation frame

- **Statement:** For social methods, transfer proxies align more strongly than trading-success proxies on this chain.
- **Evidence:** Table 4.7: transfer means 0.333 (ER) and 0.534 (AWP) exceed trading-success means 0.096 (ER) and 0.179 (AWP).

8.5.5 SRQ1 / SC1 — GF-PR ceiling

- **Statement:** GF-PR bounds trading-outcome alignment via construct overlap.
- **Evidence:** Table 4.7 row GF-PR: inverse-risk 0.376 , trading success 0.583 ; Section 4.7.

8.5.6 Claims C1–C4

- **C1:** runtime evidence as in H3/RQ3.
- **C2:** credible alignment on selected proxies—allowance for ER, transfer for AWP—in Tables 4.9 and 4.8.
- **C3:** construct-specific winners as in H2/RQ2.
- **C4:** efficiency–alignment trade-off combining C1 with allowance $\tau = 0.355$ and transfer $\tau = 0.333$ for EndorseRank; robustness in Tables 4.4–4.6 and Figure 4.5.

8.5.7 Explicit non-claims

The dissertation does *not* claim:

- that EndorseRank or AWP predicts trading profits from social graphs alone;
- that AWP is obsolete;
- that GF-PR is a deployable social reputation score;
- that results generalize beyond Arbitrum One, GMX V2, and the study window without replication;
- that on-chain scores replace collateral or legal identity systems.

8.5.8 Artifact checklist for replication

1. Merged wallet-ranking table — scores for ER, AWP, GF-PR.
2. Decoded GMX close events with PnL and liquidation flags.
3. Machine-readable evaluation summary — authoritative numerics.
4. Exported LaTeX table fragments.
5. Exported vector figures.
6. Optional archived extended-baseline matrices.

Replicators can regenerate tables and figures by running the evaluation pipeline on real data with

robustness sweeps and table export, then regenerating figures from the same artifacts.